Contribution Statement: Theory DA3

Intelligent Wi-Fi Relay Drone for Autonomous Disaster Communication | LOKESHWARAN P, 23BAI1098

Base paper: Schulman et al., Proximal Policy Optimization Algorithms, 2017. It gives a stable general algorithm, but assumes a task with a defined reward and clean observations. This work turns it into a working multi-drone mission.

What is new in this work

1. Mapless navigation reward. Progress is measured along a Dijkstra shortest-path field that the policy never sees, so flying around walls pays and pushing into them does not. The policy flies unseen streets from 24 range rays alone.

2. Landing reward redesign. The first reward had a local optimum: a timeout cost less than a failed landing, so the policy learned to hover forever. Equal penalties plus potential-based shaping, $\Phi = -0.1(\text{altitude} + 2 \cdot \text{offset})$, removed it.

3. Training on the real detector's measured errors. Missed and false detections, position error and dropped frames were measured from the actual U-Net and used as training noise, so the policy learns not to chase bad detections.

4. Curriculum over disaster severity. Town difficulty (rubble, collapses, wind) for navigation, and gusts from 0 to 4.5 m/s for landing, advance automatically on smoothed reward.

5. Hierarchical multi-drone mission. A radio-aware planner (frontier exploration, landing-spot rule, shared map) directs the two PPO skills, so drones land one by one into a relay chain.

Measured effect (held-out towns, 100 runs per condition)

Contribution	Without	With
Landing reward redesign (training success)	0 %	100 %
Measured-noise training (touchdown error, 0–6 m/s gusts)	3–7 cm	1–5 cm
Landing at 6 m/s gusts vs PID visual servoing	44 %	100 %
Landing at 3 m/s gusts, 2× detector noise, vs PID	0 %	100 %
Navigation, hardest towns, vs potential field (same sensors)	54 %	80 %
Missions reaching the camp: learned vs scripted navigation	9 %	47 %
Missions reaching the camp: spot rule vs random spot	25 %	47 %
Shared map: exploration time for later drones	52 s	30 s

Final system: with the real detector in the loop, the camp is covered in 45 % of 100 unseen level-2 towns (51 % with perfect vision), and coverage extends 62 m toward it on average.

Limits

All results are from simulation. Mission success is limited mainly by whether a landable spot exists near each corner within radio range. Policies trained on a fast flight model drop from 47 % to 30 % mission success on the full quadrotor. The curriculum and path-progress reward were not tested in isolation, so no separate figure is claimed for them.